Algebra Monsoon 2026 Lecture 3

Quotients

Def Let M be a magma. A <u>congruence</u> on M is a subset $R \subset M \times M$ such that

1) R is an equivalence relation on M
2) R is a submonoid of $M \times M$

Explicitly, 1) means that

1.1) $(x,x) \in R$ for all $x \in M$,

1.2) $(x,y) \in R \Rightarrow (y,x) \in R$ for all $x, y \in M$, and

1.3) $(x,y), (y,z) \in R \Rightarrow (x,z) \in R$ for all $x, y, z \in M$.

2) means that

$(x_1, y_1), (x_2, y_2) \in R$

$\Rightarrow (x_1 x_2, y_1 y_2) \in R$.

One often writes $x \sim_R y$ or simply $x \sim y$ for $(x, y) \in R$.

The conditions then become

1.1bis) $x \sim_R x$ for all $x \in M$

1.2bis) if $x \sim_R y$, then $y \sim_R x$ for all $x, y \in M$

1.3bis) if $x \sim_R y$ and $y \sim_R z$, then $x \sim_R z$ for all $x, y, z \in M$.

which express the fact that $\sim_R$ is an equivalence relation, and

2bis) if $x_1 \sim_R y_1$ and $x_2 \sim_R y_2$,

then $x_1 x_2 \sim_R y_1 y_2$

for all $x_1, y_1, x_2, y_2 \in M$.

which expresses that fact that

the equivalence relation $\sim_R$

is <u>compatible</u> with the law of

composition in M

Example. Fix an integer $n \in \mathbb{Z}$. In the additive module $\mathbb{Z}$, the relation

$$a \equiv_n b \iff a - b \in n\mathbb{Z}$$

is a congruence (here $n\mathbb{Z}$ is the set of all multiples of n). Explicitly, if $a \equiv_n a'$ and $b \equiv_n b'$, then

$$a + b \equiv_n a' + b'.$$

If $a \equiv_n b$, we say that a is congruent to b modulo n and write $a \equiv b \pmod{n}$.

- On the additive magma $\mathbb{Z}$, the eq. rel. $x \sim y \iff (x = y \text{ or } x = -y)$ is not a congruence.

Example Let $\pi : M \to N$ be a morphism of magmas. The relation $\sim$ on M defined by $x \sim y \iff \pi(x) = \pi(y)$ is a congruence. Indeed, $\sim$ is clearly an equivalence relation. It is a congruence because

$$
\begin{aligned}
&(x_1 \sim y_1 \text{ and } x_2 \sim y_2) \\
\Rightarrow\ &\big(\pi(x_1) = \pi(y_1) \text{ and } \pi(x_2) = \pi(y_2)\big) \\
\Rightarrow\ &\pi(x_1)\pi(x_2) = \pi(y_1)\pi(y_2) \\
\Rightarrow\ &\pi(x_1 x_2) = \pi(y_1 y_2) \\
\Rightarrow\ &x_1 x_2 \sim y_1 y_2
\end{aligned}
$$

□

There is a converse:

Proposition Let M be a magma, R a congruence on M, $\overline{M} = M/R$ the set of R-equivalence classes, and $\pi : x \longmapsto \overline{x}$ the canonical surjection $M \longrightarrow \overline{M}$ (so that $\overline{x} = \langle x \rangle_R$ is the R-equivalence class of x). There is a unique law of composition

$$\overline{M} \times \overline{M} \longrightarrow \overline{M}$$

such that π becomes a morphism of magmas.

For $\overline{x}, \overline{y} \in \overline{M}$,

of M/R

Proof We need to show that for $\bar{x}, \bar{y} \in \overline{M}$, the R-equivalence class $\overline{xy}$ depends only on $\bar{x}$ and $\bar{y}$ (and not on the choice of $x \in \bar{x}$ and $y \in \bar{y}$). In other words, we have to show that if $x' \in \bar{x}$ and $y' \in \bar{y}$, then $\overline{x'y'} = \overline{xy}$. But this follows from the fact that R is compatible with the law of composition in M:

$$(x' \in \bar{x} \text{ and } y' \in \bar{y})$$
$$\Rightarrow (x' \sim_R x \text{ and } y' \sim_R y)$$
$$\Rightarrow x'y' \sim_R xy$$
$$\Rightarrow \overline{x'y'} = \overline{xy}.$$

Therefore there is a unique mapping

$$\overline{M} \times \overline{M} \longrightarrow \overline{M}$$

such that $(\overline{x}, \overline{y}) \longmapsto \overline{xy}$.

With this law of composition on $\overline{M}$,
the canonical surjection $x \longmapsto \overline{x}$
is obviously a morphism of magmas $M \to \overline{M}$. □

Def We say that the magma $\overline{M} = M/R$ is the quotient of the magma M by the congruence R. (More precisely, it is the morphism $M \to M/R$ which is the quotient of M by R.)

Proposition (Universal property of the quotient)

Let M be a magma, R a congruence on M, M/R the quotient magma, and $\pi : M \to M/R$ the canonical surjection.

For every morphism of magmas $f : M \to N$ such that

$$x \sim_R y \implies f(x) = f(y),$$

there exists a unique morphism of magmas $\overline{f} : M/R \to N$ such that $f = \overline{f} \circ \pi$. (this condition is expressed by saying that the triangle commutes).



Proof As M/R is the quotient of the set M by the equivalence relation R, there is a unique mapping $\overline{f} : M/R \to N$ such that $f = \overline{f} \circ \pi$. It remains to show that $\overline{f}$ is a morphism of magmas. Indeed, π is surjective and for $x, y \in M$

$$\begin{aligned}
\overline{f}(\pi(x)\pi(y)) &= \overline{f}(\pi(xy)) \\
&= f(xy) \\
&= f(x) f(y) \\
&= \overline{f}(\pi(x))\, \overline{f}(\pi(y)),
\end{aligned}$$

showing that $\overline{f}$ is a morphism. □

Let $f: M \to N$ be a morphism of magmas and take R to be the congruence on M defined by

$$x \sim_R y \iff f(x) = f(y).$$

We have seen in Sets and Mappings that $\overline{f}$ is a bijection onto $f_*(M)$. We have also seen that a bijective morphism of magmas is an isomorphism. We thus have a canonical decomposition of f:

$$M \xrightarrow{\pi} M/R \xrightarrow{\overline{f}} f_*(M) \xrightarrow{i} N$$

where i is the inclusion. So every morphism is a surjective morphism followed by an isomorphism followed by an injective morphism.

Proposition Let M be a magma, R a congruence on M and $\pi : M \to M/R$ the quotient of M by R. If M is a monoid (resp. group, resp. module) then so is M/R and π is a morphism of monoids (resp. groups, resp. modules).

Proof This follows from a previous proposition because π is surjective. □

Example Fix $n \in \mathbb{Z}$. We denote the quotient of the additive module $\mathbb{Z}$ by the congruence $\equiv_n$ by $\mathbb{Z}/n\mathbb{Z}$.

Exercise Show that the equivalence relation $\equiv_n$ is compatible with the multiplication in $\mathbb{Z}$ (meaning, if $x_1 \equiv_n y_1$ and $x_2 \equiv_n y_2$, then $x_1 x_2 \equiv_n y_1 y_2$).